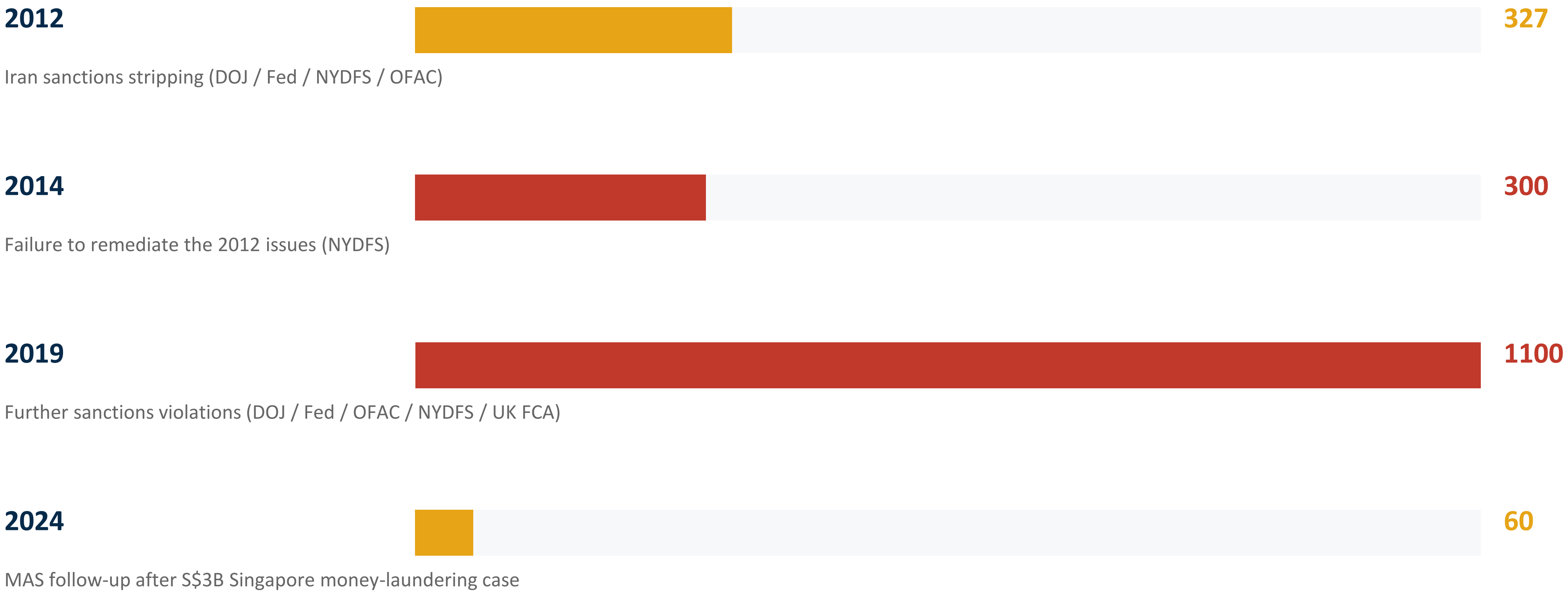
Strait Compliance

Jurisdiction-aware AML monitoring built for banks
that clear in two regulators at once.

Vivin Joghee · G2505378K · NTU MH6822 · Assignment 1 (Option A: Working Prototype)

Why we exist

Standard Chartered: ~USD 1.8B in AML penalties over 12 years



The 2014 penalty was for FAILURE to remediate the 2012 issue — same monitoring system, second offence.

Why this product is right for our company

US NY Branch	**Federally licensed; supervised by Fed + NYDFS. Clears USD through CHIPS / Fedwire.**
SG SG Subsidiary	Locally incorporated under Singapore Banking Act §7; supervised by MAS.
💵 Every USD wire from SG	Touches US OFAC scope via NY correspondent clearing. No exceptions.

Every cross-border SGD→USD flow is subject to two regulators with different rules.

A monitoring system built for one regulator silently violates the other every day. This is the exact failure mode of 2012 and 2019.

What a memo or spreadsheet cannot do

Action	Manual (spreadsheet)	Strait Compliance
Transactions reviewed	~50 / day per analyst	1,000+ / second
Rules applied per transaction	1 jurisdiction at a time	2 jurisdictions in parallel
Citation trail per alert	Free-text comment	rule_id + rule_pack_sha + model_sha
Reproducible after rule change?	No	Yes — SHA history
Alerts in our prototype run	—	1,250 alerts on 1,000 transactions
Contradictions surfaced	Invisible	78 explicit, queued for human review

The spreadsheet collapses under volume; the system pins every alert to a regulation citation that survives examination.

What regulatory divergence we handle

9 dimensions where US and Singapore disagree — each encoded as its own YAML rule with SHA-256

Dimension	United States	Singapore
Primary statute	BSA / PATRIOT / AMLA 2020	CDSA / TSOFA / MAS Notice 626
Lead administrator	FinCEN + 4 prudential + OFAC	MAS (single integrated regulator)
SAR/STR floor	≥ USD 5,000 + suspicion (30-day filing)	No floor — suspicion alone (CDSA s.45)
Cash transaction report	> USD 10,000 (31 CFR §1010.311)	No general bank threshold
PEP scope	Foreign only (31 CFR §1010.605)	Broader — foreign + domestic + IO
Sanctions list	OFAC SDN (extraterritorial via USD)	MAS-TFS (no extraterritorial reach)
Beneficial ownership	CTA / BOI — paused 2025	ACRA Register of Registrable Controllers
AI governance	OCC 2026-13 (excludes GenAI)	MAS FEAT + Veritas (covers AI fully)
2026 direction	Mixed	Tightening

How the divergence shows up in real data

1,000 synthetic transactions → 78 contradictions across all four shapes

US flag / SG block

MAS principles-based block where US only flags

US block / SG flag

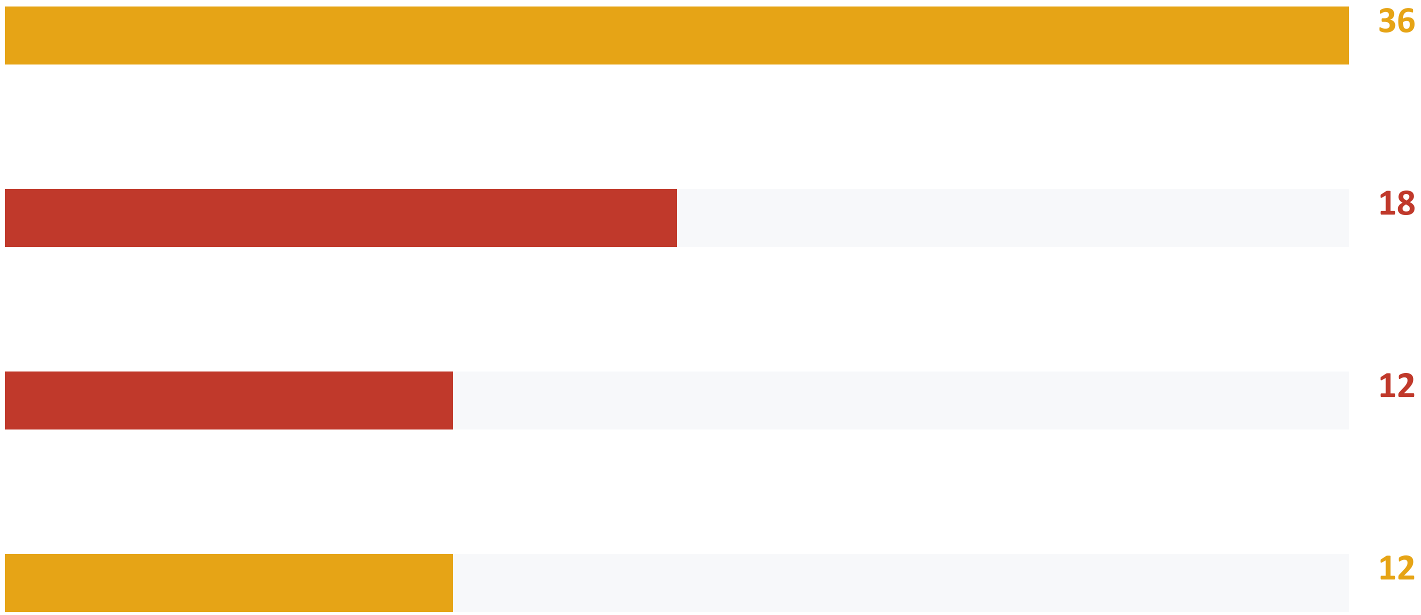
OFAC SDN + co-firing foreign PEP

US block / SG allow

OFAC SDN, weak SG signal

US allow / SG block

SG TBML where US has no rule lever



The system never silently picks a side.

Every contradiction lands in a pending queue for the MLRO to resolve. The drawer explains the regulatory pattern in plain English.

The audit primitives a regulator expects

Every alert in our database carries:

Column	Purpose
transaction_id	Joins to the underlying flow
rule_id	e.g. US.SANCTIONS.OFAC_SDN — cite-able to a CFR section
rule_pack_sha (256)	e.g. 7c1f8093964bfcb8… — exact YAML version
model_sha (256)	e.g. e8b7613118fde18a… — exact model weights
ml_score + shap_attribution	Top-12 features that drove the score

Model performance (held-out test, threshold tuned only on validation):

ROC-AUC	PR-AUC	F1-optimal threshold	Sanctions recall	Large-cash recall
0.97	0.93	0.5960	100%	100%

What we don't do — boundary choices, not gaps

We don't	Why
KYC onboarding (biometric, doc verification)	Specialist vendors (Onfido / Sumsub) do this better. Out of scope.
LLM-drafted SAR/STR narratives	OCC Bulletin 2026-13 disclaims governance over GenAI. Liability stays with us.
EU 6AMLD compliance	Not properly studied. Pretending otherwise is the failure mode this tool prevents.
Real-time blocking decisions	Bank's payment switch makes the block call; we supply the signal + SHA-pinned reason.
Fictional sanctioned entities (real names)	Synthetic flags only via raw columns; we never claim a real entity is sanctioned.

Each “no” is a defended choice, not a roadmap item. The discipline is the product.

Live and reproducible.

Web app: https://strait-compliance.pages.dev

Source: https://github.com/vivin-joghee/regtech-tool

Strait Compliance does not promise to fight financial crime.
It promises that when our bank's two regulators disagree, the system says so to a human,
and the audit trail is reproducible six months later.